## Blessed G/Michael Catholic Secondary School

## First Semester Physics Final Exam for Grade 10th

**Jan, 2018E.C** **Time Allowed 1: 30 hr**

Take g = 10m/s$^2$ density of water = 1000 kg/m$^3$

**Name** Betanya Wondemagegnehu **Section** 10TH B **No** 12

### I. *Write 'True' if the statement is correct and 'False' if it is incorrect*

1. Specific gravity has **no** unit. True
2. Lightning is a natural example of electrical discharge. True
3. A body becomes positively charged when it loses electrons. True
4. The relative velocity of two bodies moving with the same velocities is zero. false
5. A body moving with constant velocity have the same instantons and average velocity. True
   a-t → change in velocity m/s² m/s
6. The area under acceleration versus time graph is equal to displacement. false
7. When mass of a given substance is doubled, its density also doubled. True $\rho = \frac{m}{V}$
8. The Young's modulus of a substance will be doubled when the stress exerted on it is doubled. True $Y = \frac{stress}{strain}$
9. When a body is charged by conduction, it always gains electrons. false
10. A neutral conductor can be charged by induction without touching a charged object. True

### II. *Choose the best answer for the following questions.*

A 11. Which one of the following statement is correct?

A) Elastic materials regain their shape after deformation.

B) Plastic materials can recover the original size or shape after being distorted.

C) Plastic materials change their shape by pressure.

D) Elastic materials cannot recover the original size or shape after being distorted.

B 12. A wire of cross-sectional area $2\times10^{-6}$ m$^2$ is stretched by a force of 50 N. The stress developed is:

A) $1\times10^7$ Pa C) $5\times10^7$ Pa

B) $2.5\times10^7$ Pa D) $1\times10^6$ Pa

D 13. The SI unit of Electric charge is

A) Ampere (A) B) Tesla (T) C) Volt (V) D) Coulomb (C)

A 14. A wire 1 m long stretches by 1 mm under a force. The strain is:

A) 0.001 C) 0.1

B) 0.01 D) 0.0001

A 15. What is the mass of a 10 L volume of water?

A) 10 kg C) 100 kg

B) 0.1 kg D) 0.01 kg

B 16. If the cross-sectional area of a wire is doubled while keeping the force constant, the stress is.

A) Doubled B) halved C) quarter D) one-fourth

$\cos 30° = sin60° = \frac{\sqrt{3}}{2}, cos37° = sin53° = 0.8, cos45° = sin45° = \frac{\sqrt{2}}{2}, cos53° = sin37° = 0.6, cos60° = sin30° = 0.5$

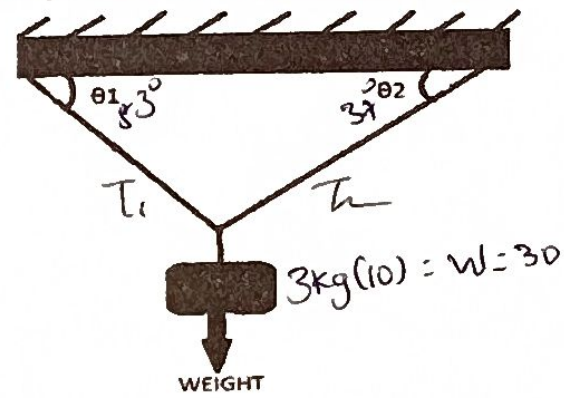
C 17. A wire 2 m long, area $1\times10^{-6}$ m$^2$ is stretched by 0.1 mm under a 100 N force. Young's modulus is:

A) $2\times10^{11}$ Pa B) $2\times10^{10}$ Pa C) $2\times10^{12}$ Pa D) $2\times10^{9}$ Pa

B 18. For the same stress, a material with higher Young's modulus will have:

A) More strain B) Less strain C) Same strain D) Cannot say

D 19. An object of mass 3kg hangs from three light inextensible wires as shown in figure. Two of the wires make angles $\theta_1=53°$ and $\theta_2=37°$ with the horizontal. If the system is in equilibrium, what will be the tensions $T_1$ and $T_2$ in the wires?




A) $T_1$= 18N and $T_2$=24 N

B) $T_1=T_2$=18 N

C) $T_1=T_2$=24 N

D) $T_1$= 24N and $T_2$=18 N

A 20. Uniformly accelerated motion means:

A) Acceleration is constant C) Displacement is constant

B) Velocity is constant D) Acceleration changes with time

D
C 21. What is the density of a 2.7 kg cube of side length 10 cm?

A) 2.7 kg/m$^3$ B) 27 kg/m$^3$ C) 270 kg/m$^3$ D) 2700 kg/m$^3$

B 22. Which one of the following laws best describes the behaviour of elastic materials?

A) Gay Lussac's law C) Newton's second law

B) Hooke's law D) Newton's law of gravitation

B 23. A steel wire ($Y=2\times10^{11}$ Pa) of length 2 m and cross-section $2\times10^{-6}$ m$^2$ is stretched by a force of 200 N. The extension is:

A) 0.5 mm B) 1 mm C) 2 mm D) 0.2 mm

D 24. The fulcrum of a uniform 20kg seesaw that is 4.0m long is located 2.5 m from one end. A 30 kg child sits on the long end. The weight of a person sitting on the other end in order to balance the seesaw is:

A. 750 N B. 850 N C. 433 N D. 567 N

B 25. A body starts from rest and accelerates uniformly at 3 m/s$^2$ for 4 seconds. What is its final velocity?

A) 7 m/s B) 12 m/s C) 9 m/s D) 15 m/s

B 26. For a free-falling object, acceleration is:

A) Increasing with time B) Constant C) Zero D) Decreasing with time

B 27. If a body starts from rest under uniform acceleration, the distance covered is:

A) Proportional to time D) Independent of time

B) Proportional to square of time C) Proportional to cube of time

A 28. Two cars A and B are moving along a straight path in the same direction with a speed of 54 km/h and 36 km/h, respectively. What is the relative velocity of car A with respect to car B?

A) 18 m/s B) 5 m/s C) 90 m/s D) 45 m/s

B 29. A car starts from rest and accelerates uniformly at 5 m/s². How long will it take to reach 50 m/s?

A) 5 s C) 15 s

B) 10 s D) 20 s

D 30. What is the purpose an electroscope?

A) To identify whether a body is charged or not.

B) To identify whether a body is a conductor or insulator.

C) To identify the type of charge in a body is positive or negative.

D) All of the above can be tested using an electroscope.

B 31. How much force is needed to stretch a 1 m wire of area $1\times10^{-6}$ m$^2$ by 1 mm if $Y=2\times10^{11}$ Pa?

A) 100 N B) 200 N C) 50 N D) 400 N

B 32. The slope of a velocity-time graph gives:

A) Displacement B) Acceleration C) Velocity D) Time

A 33. A car moving at 20 m/ slows down uniformly to rest in 5 s. What is its acceleration?

A) $-4$ m/s$^2$ C) $-3$ m/s$^2$

B) $-5$ m/s$^2$ D) $-2$ m/s$^2$

B 34. For a body moving with uniform acceleration, the velocity-time graph is:

A) A straight line passing through origin

B) A straight line with a slope equal to acceleration

C) A parabola

D) A horizontal line

A 35. A car moving at 20 m/s applies brakes and stops in 10 s. What distance does it cover before stopping?

A) 100 m B) 150 m C) 200 m D) 250 m

B 36. Which of the following quantities is dimensionless?

A) Stress B) Strain C) Young's modulus D) Force

B 37. A freely falling object starts from rest. How far will it fall in 3 seconds?

A) 30 m C) 50 m

B) 45 m D) 60 m

A 38. A body possess a (negative) charge of $-6.4$ nC. How many electrons are in excess or deficit? (Take $e = 1.6 \times 10^{-19}C$)

$-$ means excess

A) $4.0 \times 10^{10}$ electrons in excess C) $4.0 \times 10^{13}$ electrons in excess

B) $4.0 \times 10^{10}$ electrons in deficit D) $4.0 \times 10^{13}$ electrons in deficit

***III.*** **Write down the correct answer with the proper SI unit.**

A cylindrical steel wire of length L=2 m and cross-sectional area A=1 $cm^2$ is subjected to a tensile force F=10,000 N. The Young's modulus of steel is Y=$2\times10^{11}$ Pa. Find:

39. The stress in the wire : N/m 100 N/m or 1 N/m
40. The strain in the wire $0.5*10^{-3}$
41. The elongation of the wire $1*10^{-3}$

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| | | | |
|---|---|---|---|
| 1. True | 11. A | 21. D | 31. B |
| 2. True | 12. B | 22. B | 32. B |
| 3. True | 13. D | 23. B | 33. A |
| 4. ~~false~~ | 14. A | 24. D | 34. B |
| 5. True | 15. A | 25. B | 35. A |
| 6. false | 16. B | 26. B | 36. B |
| 7. ~~True~~ | 17. C | 27. B | 37. B |
| 8. ~~True~~ | 18. B | 28. ~~A~~ | 38. A |
| 9. false | 19. ~~D~~ | 29. B | 39. ~~1 N/M~~ $10^8$ Pascal |
| 10. True | 20. A | 30. D | 40. $0.5*10^{-3} = 5\times10^{-4}$ |
| | | | 41. $1\times10^{-3}$ |

35 36